

The action potential

The written notes for the action potential walkthrough. Week 4 competencies 13 to 16.

To print: these notes as a two-column PDF. Already printed the Week 4 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

The key numbers

Point	Voltage	What sets it
Rest	-70 mV	Mostly the potassium leak (resting potential walkthrough)
Threshold	about -55 mV	The inward sodium current first exceeds the outward potassium current
Peak	about +30 mV	Sodium inactivation gates close and voltage-gated potassium channels open
After-hyperpolarization	about -80 mV	Voltage-gated potassium channels are slow to close
Sodium equilibrium potential	+60 mV	The value the rising phase moves toward
Potassium equilibrium potential	-90 mV	The value the after-hyperpolarization moves toward

The whole action potential, from threshold back to rest, lasts a few milliseconds.

The two voltage-gated channels

Channel	Gates	At rest	When depolarized
Voltage-gated Na ⁺ channel	Activation gate, and inactivation gate on the cytoplasmic side	Activation gate closed, inactivation gate open	Activation gate opens quickly; inactivation gate closes a little later and stays closed until the membrane repolarizes
Voltage-gated K ⁺ channel	One gate	Closed	Opens slowly, at about the time the sodium channels inactivate, and closes slowly

- Both channels are opened by depolarization. The difference that shapes the action potential is timing: the sodium channel responds quickly and the potassium channel slowly.
- A stimulus can open activation gates, but no stimulus of any size can open an inactivation gate. It reopens only after the membrane repolarizes.

Reaching threshold

- A graded potential depolarizes the trigger zone, the axon hillock and first part of the axon, and opens some voltage-gated sodium channels.
- Below threshold (example: a 10 mV graded potential brings the membrane to -60 mV), the inward sodium current is smaller than the outward potassium current through the leak channels. More positive charge leaves than enters, so the membrane returns to -70 mV and no action potential occurs.
- At threshold (a 15 mV graded potential brings the membrane to about -55 mV), the inward sodium current equals, then exceeds, the outward potassium current.
- Threshold is defined by this balance between two currents, so its exact value can vary from neuron to neuron.

1. Sodium enters and the membrane depolarizes.
2. Depolarization opens more voltage-gated sodium channels.
3. More sodium enters, and the membrane depolarizes further.
4. Back to step 2. This cycle is positive feedback. Once it starts, the action potential no longer depends on the stimulus.

The phases

Phase	Voltage	Na ⁺ channels	K ⁺ channels	Ion movement
Rest	-70 mV	Resting (activation gate closed)	Closed	Leak channels only
Rising phase (depolarization)	-55 to about +30 mV	Open	Starting to open	Na ⁺ enters rapidly
Peak (overshoot above 0 mV)	about +30 mV	Inactivation gates close	Open	Na ⁺ entry stops; K ⁺ starts to leave
Falling phase (repolarization)	+30 back to -70 mV	Inactivated	Open	K ⁺ leaves
After-hyperpolarization	down to about -80 mV, then back to rest	Returning to the resting state: activation gate closes, inactivation gate opens	Still open, closing slowly	Extra K ⁺ leaves, moving the membrane potential toward -90 mV

- The rising phase moves toward the sodium equilibrium potential (+60 mV) because the membrane is now far more permeable to sodium than to anything else. This is the same rule as for the resting potential: the membrane potential moves toward the equilibrium potential of the ion the membrane is most permeable to.

- The voltage peaks near +30 mV, well short of +60 mV, because the sodium inactivation gates close and the voltage-gated potassium channels open.
- Potassium leaving through voltage-gated channels repolarizes the membrane. The sodium-potassium pump does not.
- The after-hyperpolarization happens because the voltage-gated potassium channels are slow to close, so for a few milliseconds the membrane is even more permeable to potassium than at rest and the membrane potential moves closer to the potassium equilibrium potential.

Threshold

The membrane potential at which the inward sodium current first exceeds the outward potassium current and the action potential begins; about -55 mV here.

Positive feedback

A cycle in which the result of a process strengthens the process itself: depolarization opens sodium channels, and the sodium that enters causes more depolarization.

Rising phase (depolarization)

The rapid rise from threshold to the peak, carried by sodium entering the cell.

Overshoot

The part of the action potential above 0 mV, when the inside is positive.

Falling phase (repolarization)

The return from the peak toward rest, carried by potassium leaving the cell.

After-hyperpolarization

The period after repolarization when the membrane potential is below the resting potential, caused by voltage-gated potassium channels that are slow to close.

How many ions move

- Only a tiny fraction of the ions cross the membrane during one action potential. Afterward, sodium inside is still about 15 mM and potassium inside is still about 150 mM.
- The concentration gradients remain after an action potential, so a neuron can fire again right away without waiting for the pump.
- The pump keeps the concentrations steady over the long term, across thousands of action potentials. It does not repolarize the membrane after each action potential.

All or none

- Below threshold there is no action potential. At or above threshold there is a full-sized action potential every time.
- Example: graded potentials of 5 and 10 mV stay below threshold. Graded potentials of 16 and 30 mV both produce action potentials that peak at about +30 mV.
- Once threshold is reached, the size of the action potential is set by the voltage-gated channels and the sodium equilibrium potential, not by the stimulus. A larger graded potential only brings the membrane to threshold a little sooner.

All or none

An action potential either occurs at full size or does not occur; its size does not depend on the size of the stimulus.

Coding stimulus strength

- Because every action potential is the same size, the size of an action potential cannot encode stimulus strength.
- Frequency coding: a stronger stimulus makes a single neuron fire more often. A larger graded potential at the trigger zone brings the membrane back to threshold sooner after each action potential. Example in the walkthrough: 2, 5 and 9 action potentials in the same period for weak, medium and strong stimuli.
- Population coding: a stronger stimulus activates more receptors, so more neurons fire. Example: the strong stimulus has four neurons firing instead of one.
- A single axon can use only frequency coding.

Frequency coding

Encoding stimulus strength by how often a neuron fires.

Population coding

Encoding stimulus strength by how many neurons fire.

Refractory periods

Period	When	Channel state	Can a second action potential start?
Absolute refractory period	From the start of the action potential until the sodium channels begin to return to the resting state	Na ⁺ inactivation gates closed	No, no matter how strong the stimulus
Relative refractory period	During the after-hyperpolarization	Some Na ⁺ channels have returned to the resting state; extra K ⁺ channels still open	Yes, but only with a stronger stimulus than usual, and the second action potential is often a little smaller

- Example: during the after-hyperpolarization the membrane is near -78 mV. The same 16 mV stimulus that fired the first action potential brings it only to about -62 mV, below threshold.
- The absolute refractory period lasts about 1 to 2 ms, which sets an upper limit on firing rate: $1,000 \text{ ms} \div 2 \text{ ms} = 500 \text{ per second}$, or $1,000 \text{ ms} \div 1 \text{ ms} = 1,000 \text{ per second}$. Most neurons fire well below this.
- The refractory period also keeps the action potential traveling in one direction along the axon (conduction walkthrough).

Absolute refractory period

The time during which no stimulus of any size can start a second action potential, because the sodium inactivation gates are closed.

Relative refractory period

The time that follows, during the after-hyperpolarization, when a second action potential can start but only with a stronger stimulus than usual.

Patient: a drug that blocks potassium channels

- Dalfampridine (4-aminopyridine) blocks voltage-gated potassium channels. It is used to help people with multiple sclerosis walk.

- In multiple sclerosis, axons lose myelin, which exposes potassium channels that are normally covered by it.
- With the voltage-gated potassium channels blocked, the rising phase and peak barely change, because they depend on sodium. The falling phase is much slower; the membrane still repolarizes as the sodium channels inactivate and potassium leaves through the leak channels. The action potential becomes wider, and the after-hyperpolarization mostly disappears.
- A wider action potential delivers more charge to the next segment of axon, which helps the signal cross a segment that has lost its myelin.
- Blocking these channels throughout the brain can also make neurons more excitable, so the dose is kept low: too much can cause seizures.